

Quiz 6 Class 1 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**F**) Deontological ethical framework judges an action based strictly on its resulting consequences and whether it maximizes collective happiness. (True/False, 2 pts)
2. (**D**) Which is **FALSE** about cache timing attacks? (2 pts)
 - A. They and TENEX password attack both belong to side channel timing attacks.
 - B. They exploited the fact that memory contents in cache can be accessed much faster.
 - C. They don't need a software bug like buffer overflow or memory corruption to work.
 - D. They can't work if the attackers cannot access memory beyond their privilege level.
3. (**C**) In symbolic execution, what is an SMT (Satisfiability Modulo Theories) solver responsible for? (2 pts)
 - A. Compile source codes into executables that can accept symbolic execution.
 - B. Examine branches of an if condition and decide which path to take.
 - C. Analyze a set of path conditions and determine if a specific path is feasible.
 - D. Optimize the search algorithm so each execution can cover more code.
4. (Short Answer, 4 pts) Perform symbolic execution for the source code below to determine the exact path condition required to execute the `abort()`; statement on line 7. Provide concrete values of a and b that would satisfy the condition, one each.


```

1  int a = read_input(); // Symbolic value alpha
2  int b = read_input(); // Symbolic value beta
3  int c = a - 7;
4  if (c < 3) {
5      b = b + c;
6      if (b > 8) {
7          abort(); // Target Line
8      }
9  }
```

Solution:

Line 4 $c < 3 \rightarrow a - 7 < 3 \rightarrow a < 10$

Line 6 $b > 8 \rightarrow b + c > 8 \rightarrow b + a - 7 > 8 \rightarrow b + a > 15$

Final path condition for line 7: $a < 10 \ \&\& \ a + b > 15$

One possible pair for a and b: $a == 9, b == 7$